

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – I

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Class:TYIT	Batch:1
Date of Assignment:24/7/26	Date/Time of Submission:24/7/26

Part A - Theory:

a) Connect to a CSV Dataset and Explore the Tableau Interface

Tableau allows users to connect to CSV files for data analysis. After loading the dataset, users can explore the Data pane, Sheets, Rows, Columns, Marks card, and Toolbar to create visualizations efficiently.

b) Identify and Differentiate Measures and Dimensions

Dimensions are qualitative fields (e.g., Name, Category, Region) used for grouping and categorizing data. Measures are quantitative fields (e.g., Sales, Profit, Quantity) used for calculations and analysis.

c) Convert Fields Between Discrete and Continuous

In Tableau, fields can be displayed as **Discrete (blue)** or **Continuous (green)**. Discrete fields create headers and separate categories, while continuous fields create axes for numerical analysis, changing the visualization style.

d) Create and Customize a Basic Bar Chart

A bar chart is used to compare values across different categories. Users can customize the chart by changing colors, sorting data, adding labels, adjusting sizes, and formatting axes for better readability.

e) Build a Line Chart for Time-Series Analysis

A line chart displays trends over time by connecting data points with lines. It is useful for analyzing changes, patterns, and growth across dates or time periods.

f) Create a Geographic Map Visualization

Tableau automatically recognizes geographic fields such as Country, State, or City and plots them on a map. Maps help visualize regional patterns, distributions, and location-based insights.

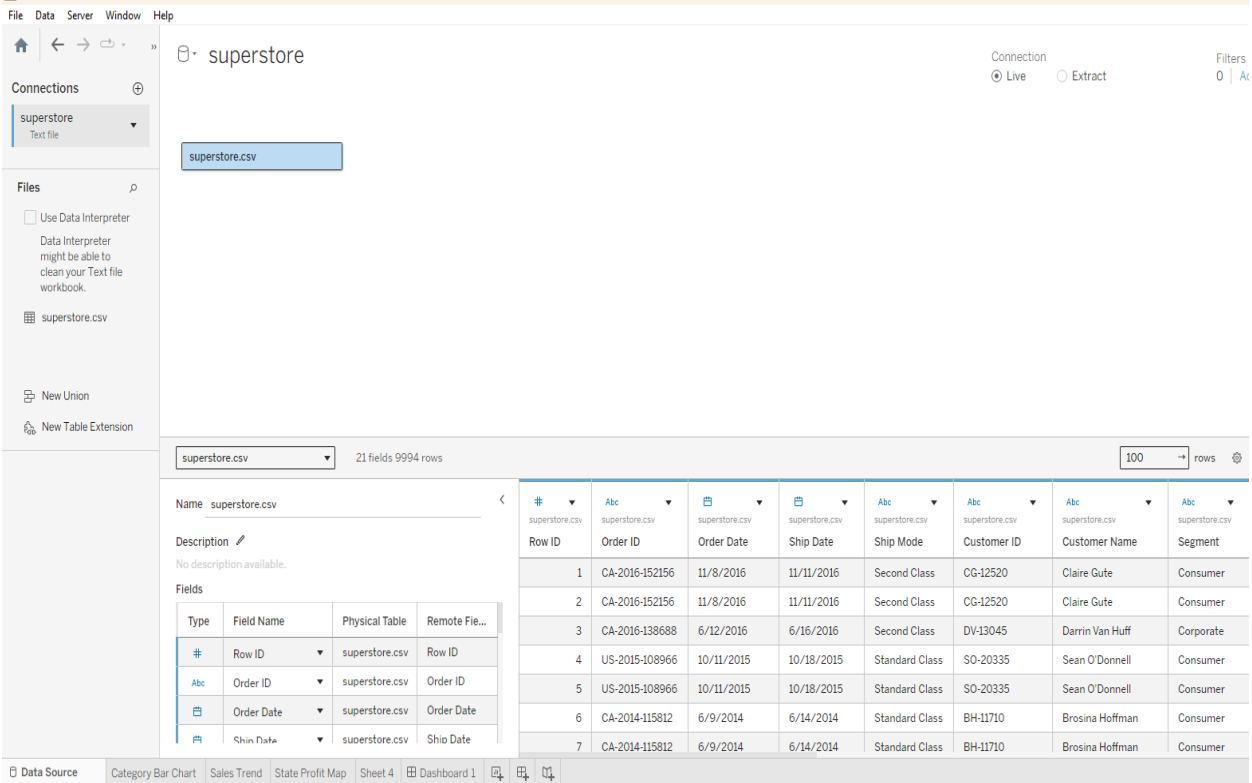
g) Combine Multiple Charts into a Simple Interactive Dashboard

A dashboard combines multiple worksheets into a single interactive view. Filters, actions, and highlights can be added so users can explore and compare data across different visualizations in one place.

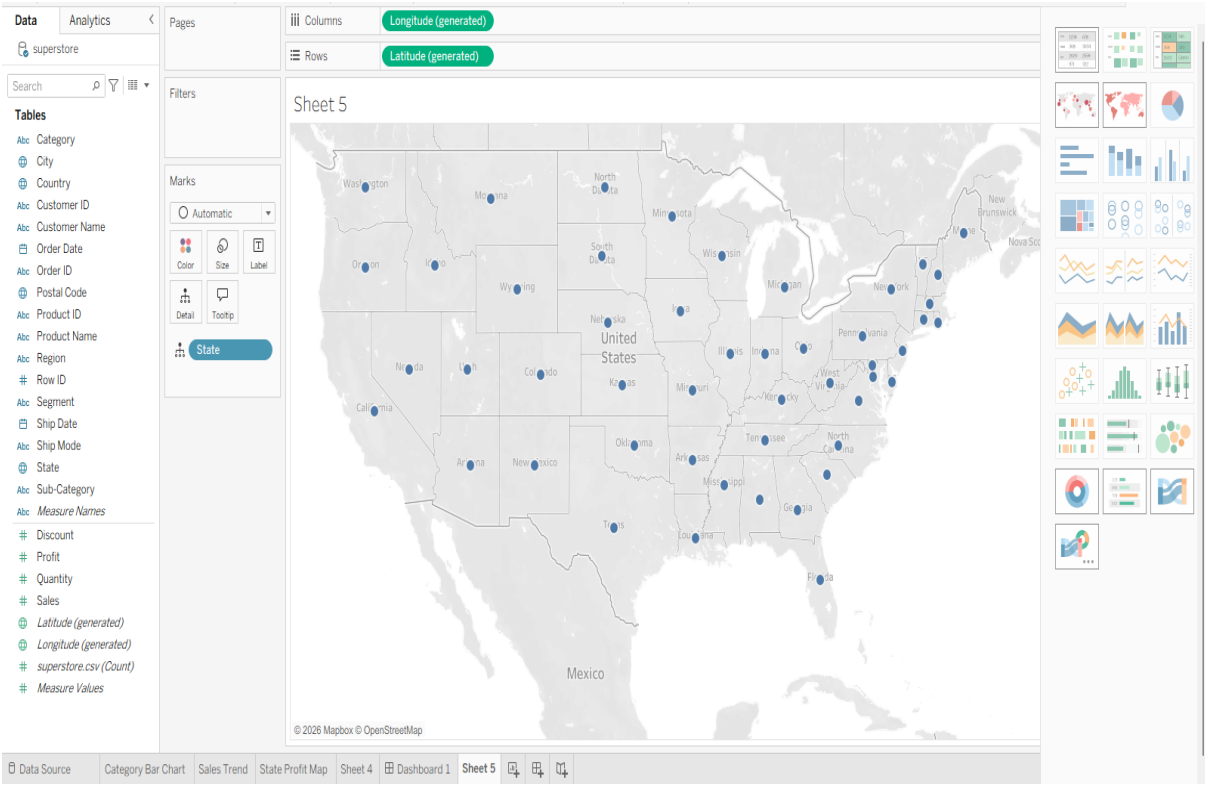
Part B – Program

Getting Started with Tableau and Building Basic Visualizations

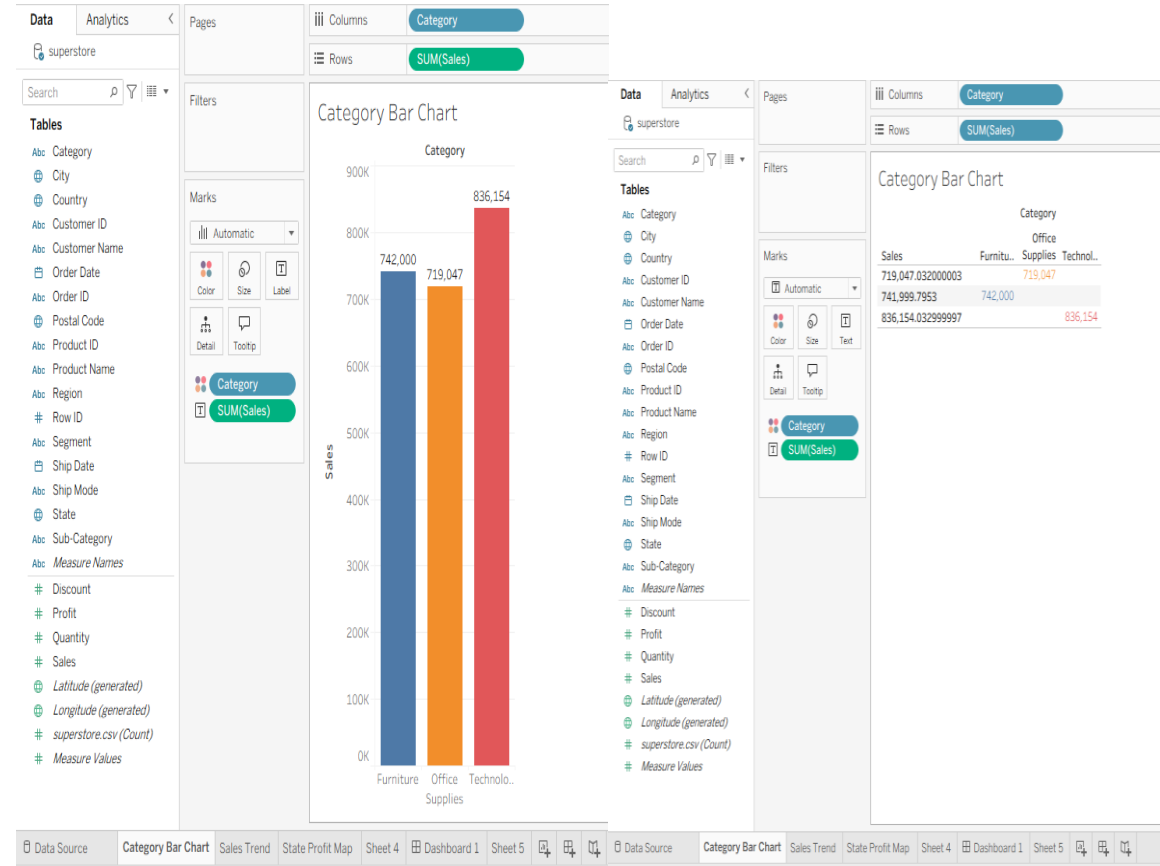
a) Connect to a CSV dataset and explore the Tableau interface



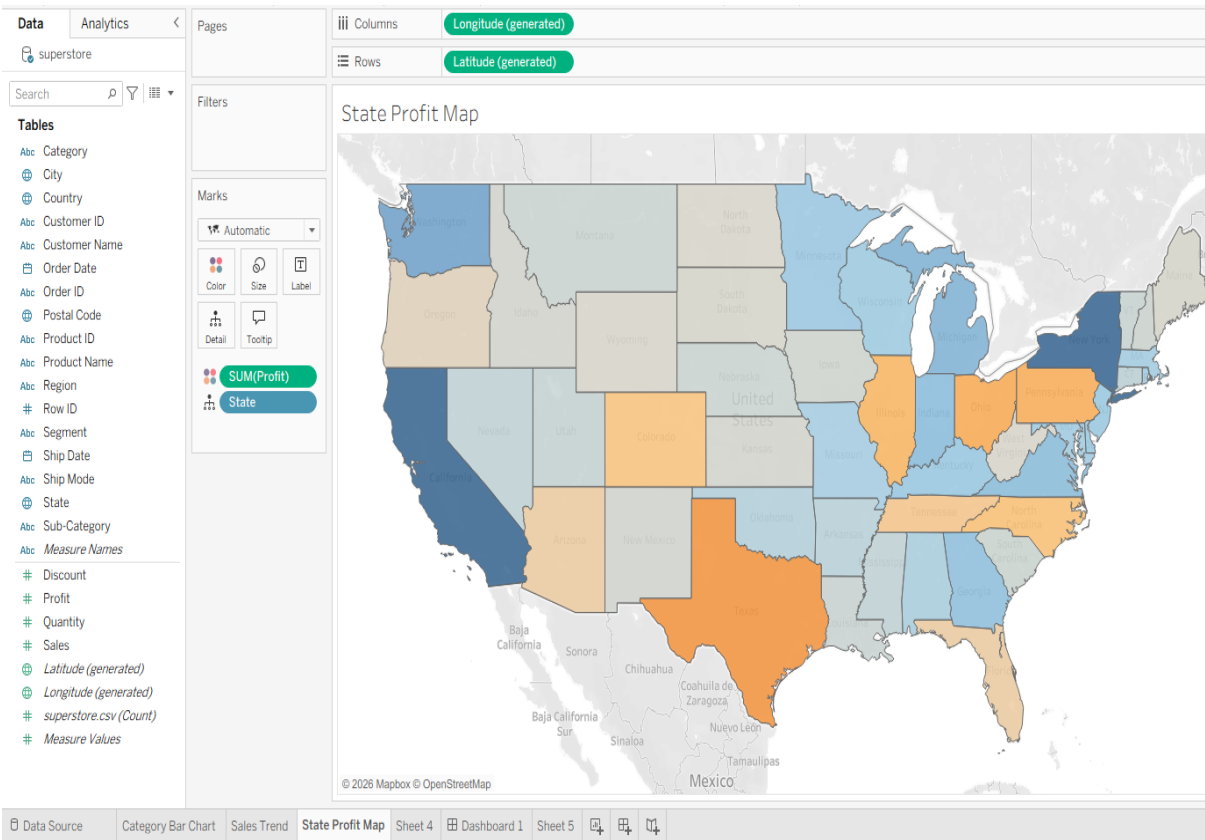
b) Identify and differentiate Measures and Dimensions



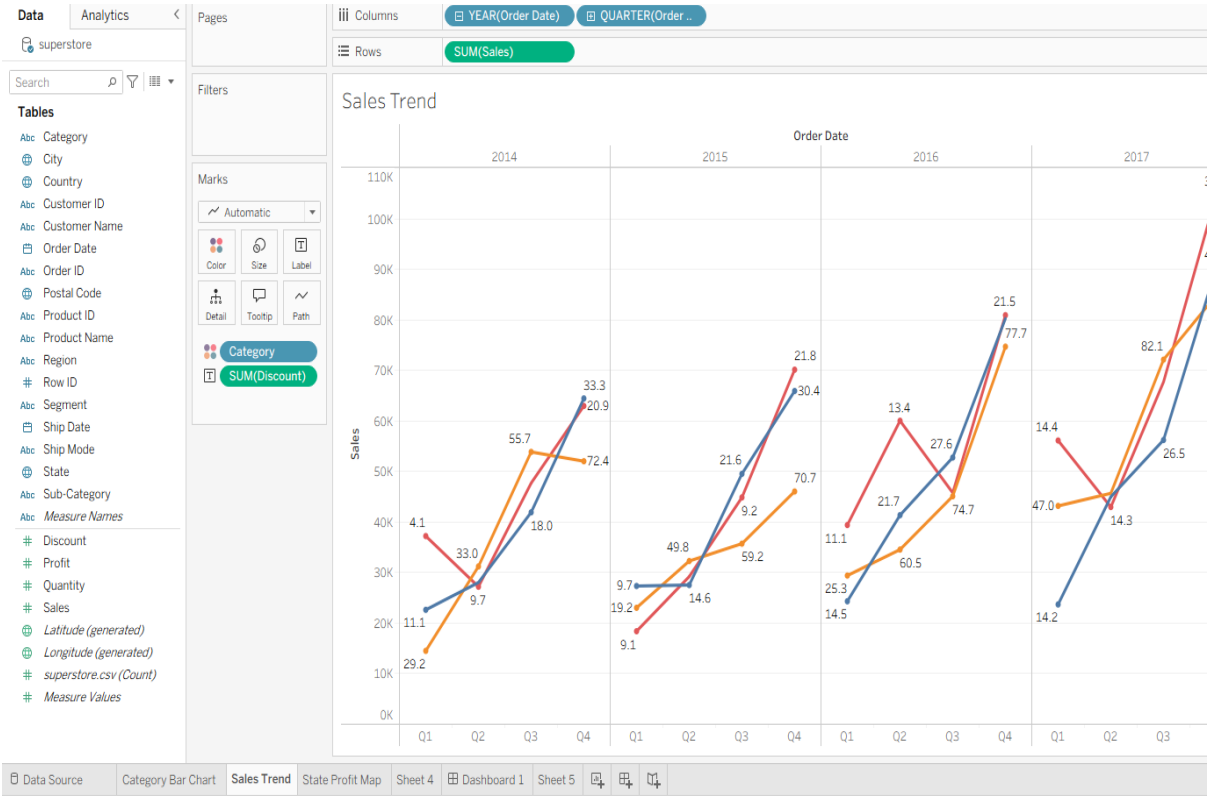
c) Convert fields between discrete and continuous and observe changes



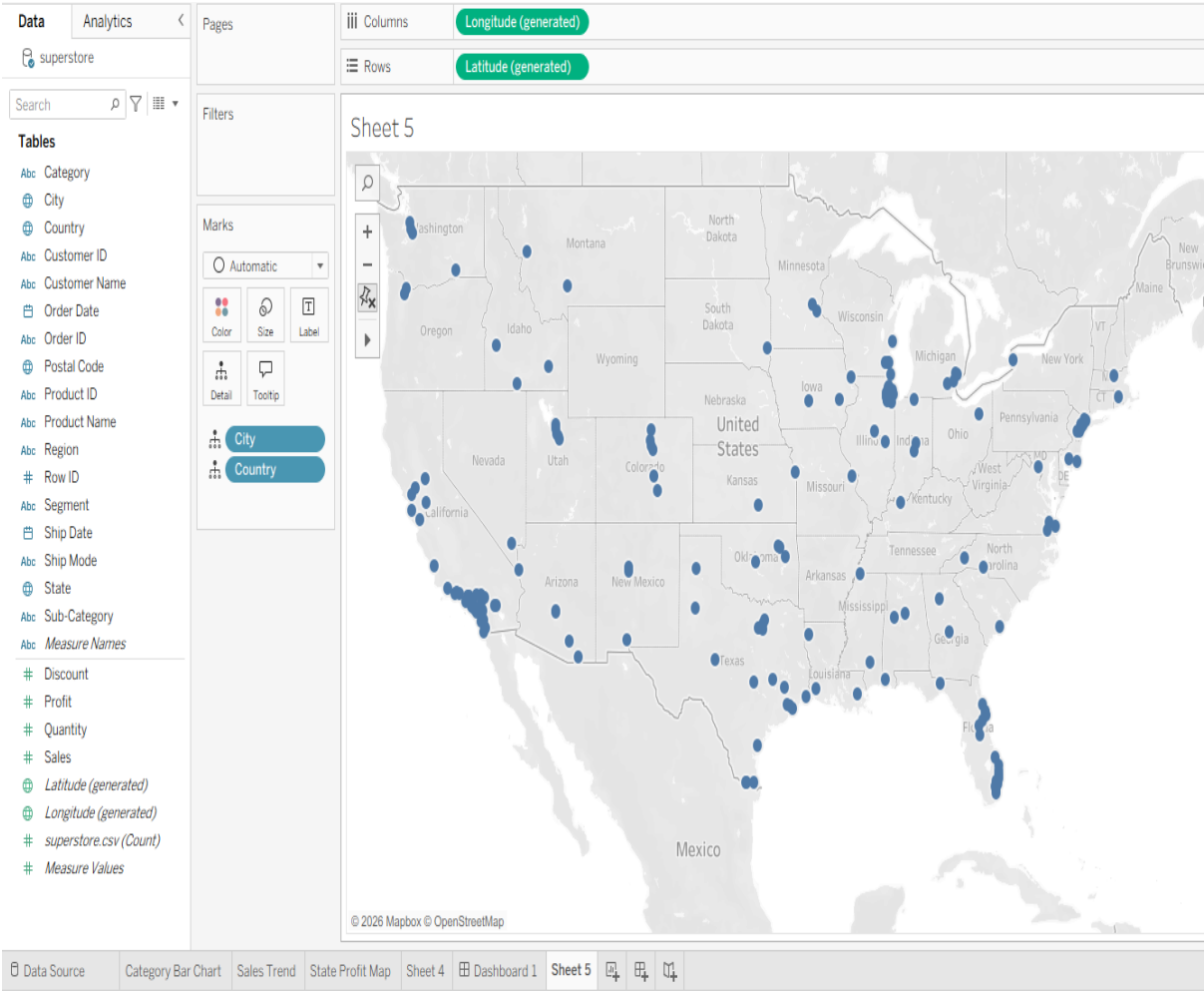
d) Create and customize a basic bar chart



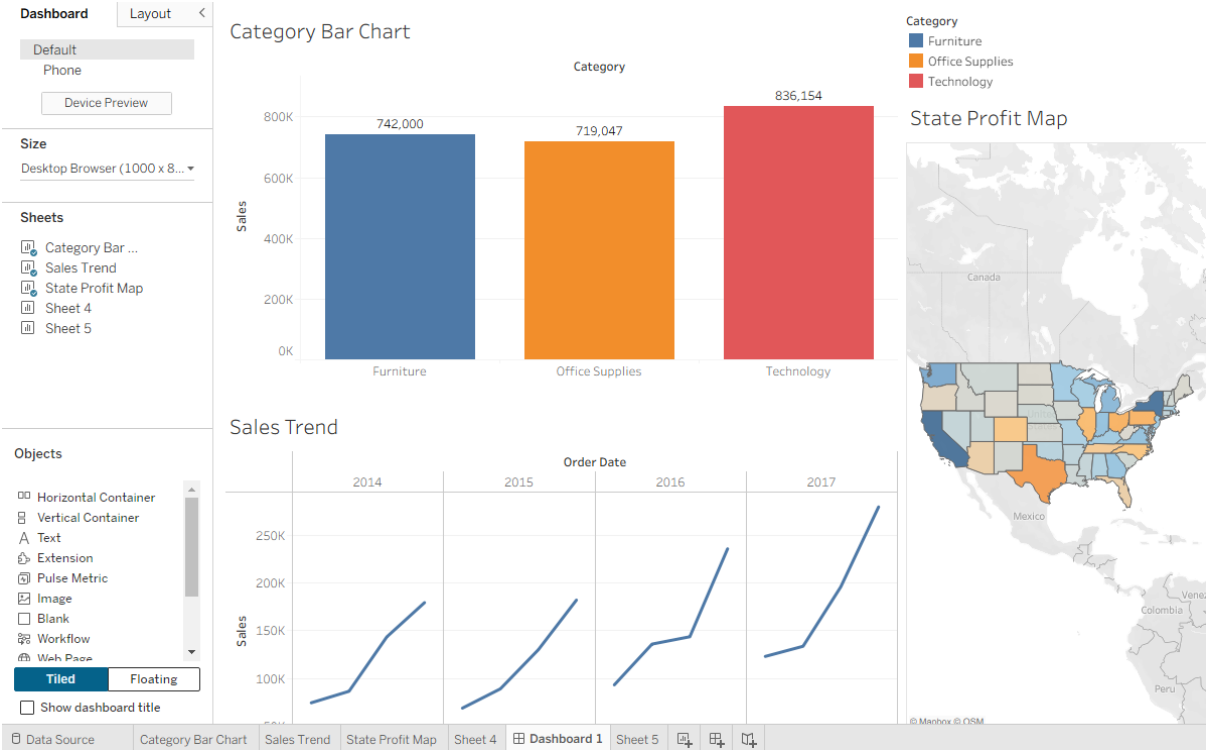
e) Build a line chart for time-series analysis



f) Create a geographic map visualization



g) Combine multiple charts into a simple interactive dashboard



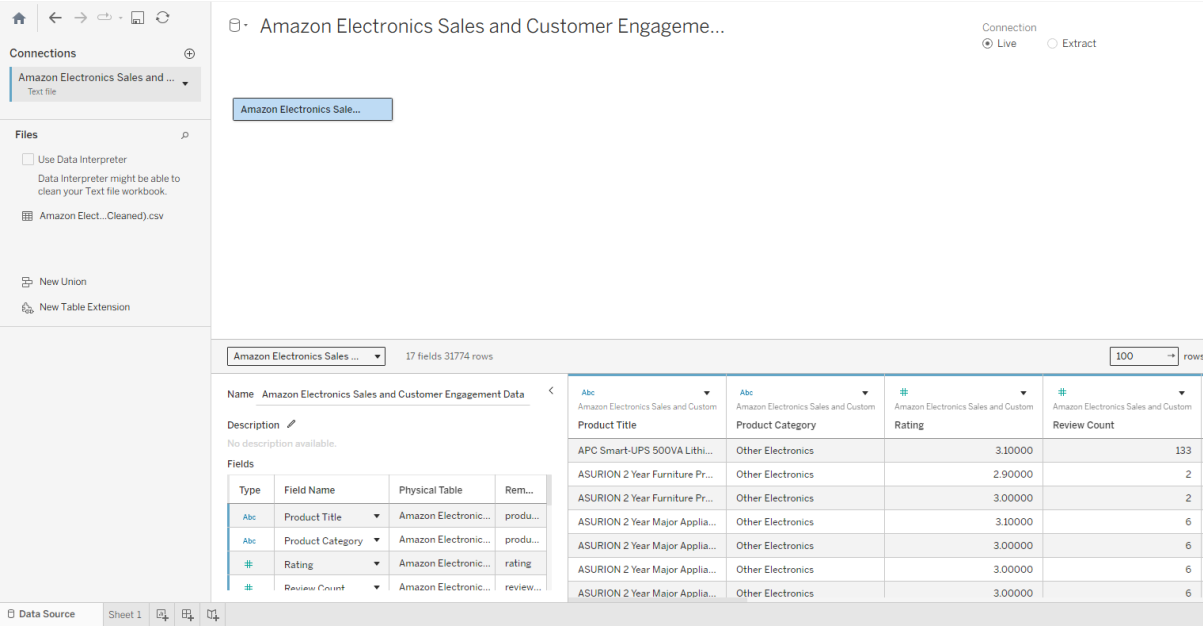
Part C – Curiosity Question

Curiosity Question

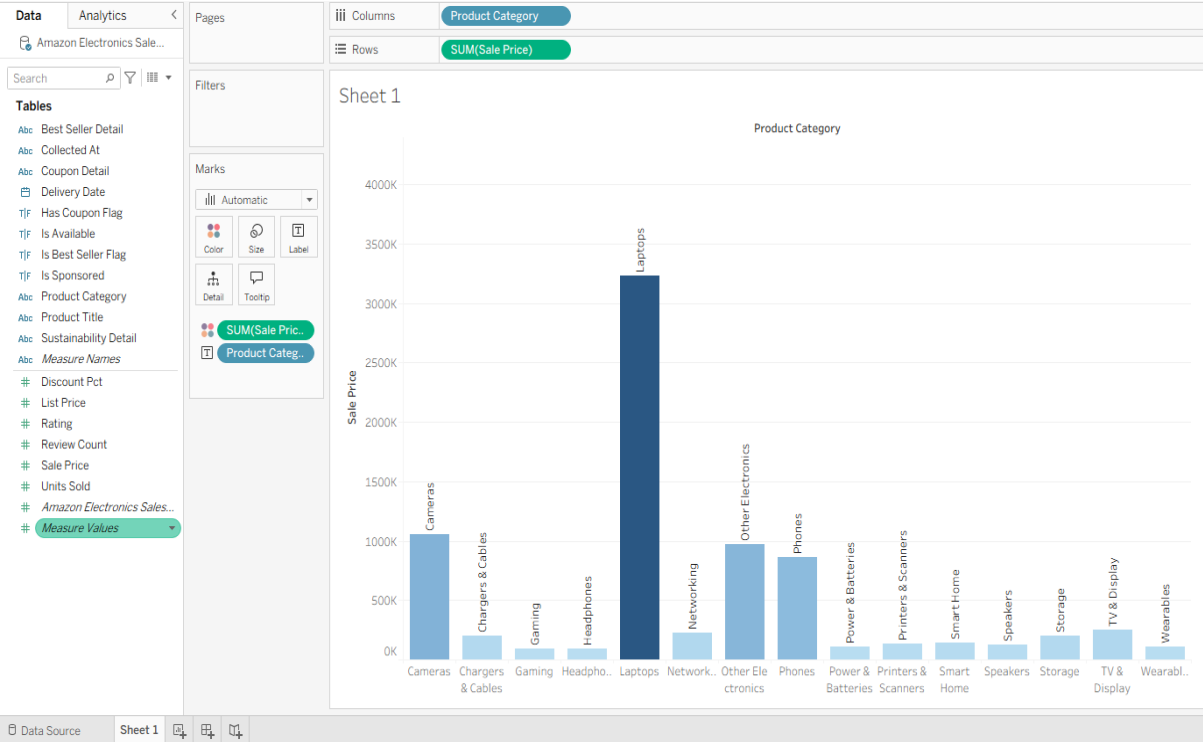
Question:
Using a dataset of your choice (CSV format), create an interactive Tableau dashboard that effectively presents key insights from the data.

Requirements:

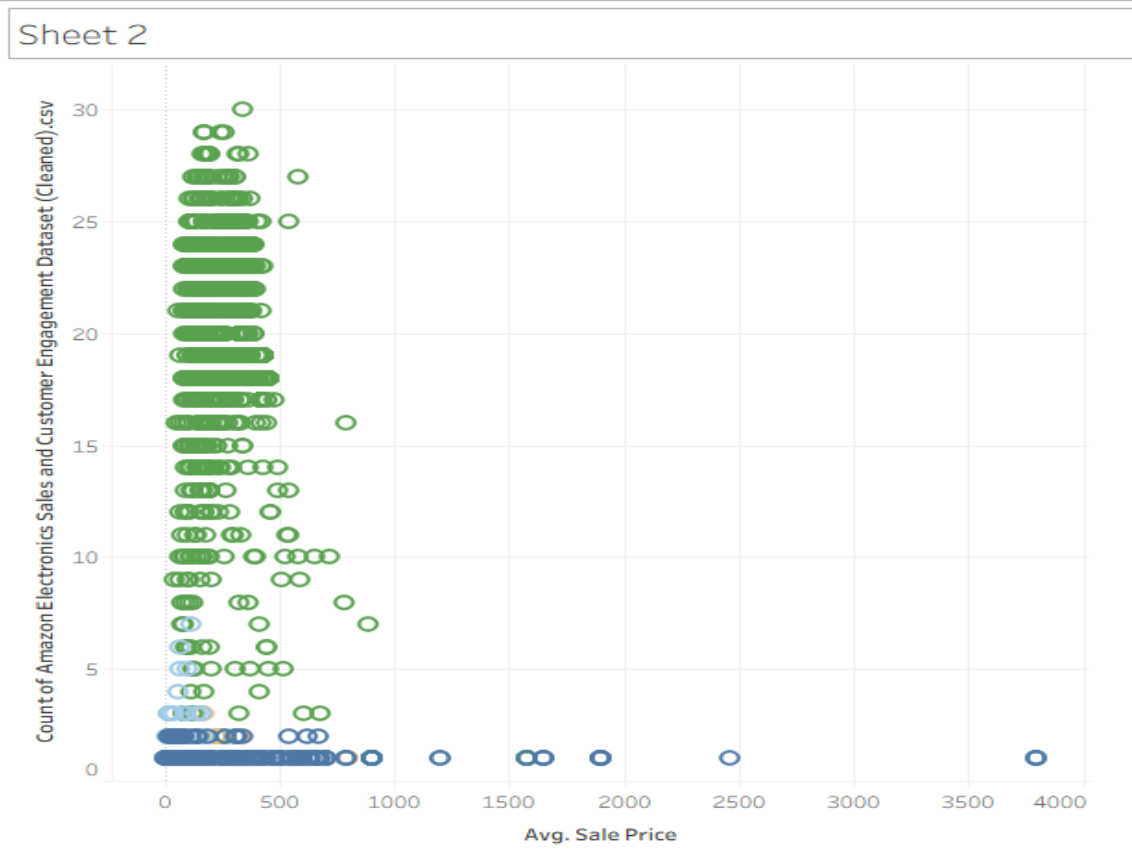
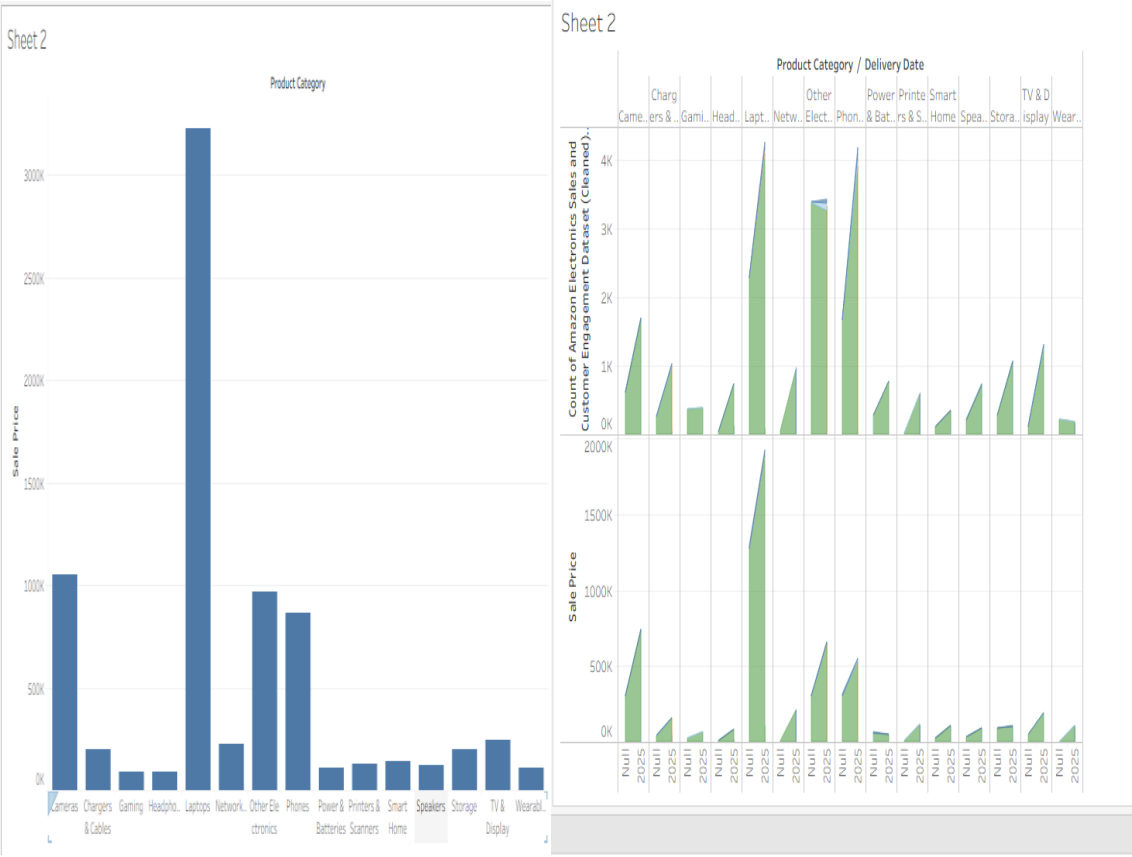
- 1. Import your own dataset into Tableau.



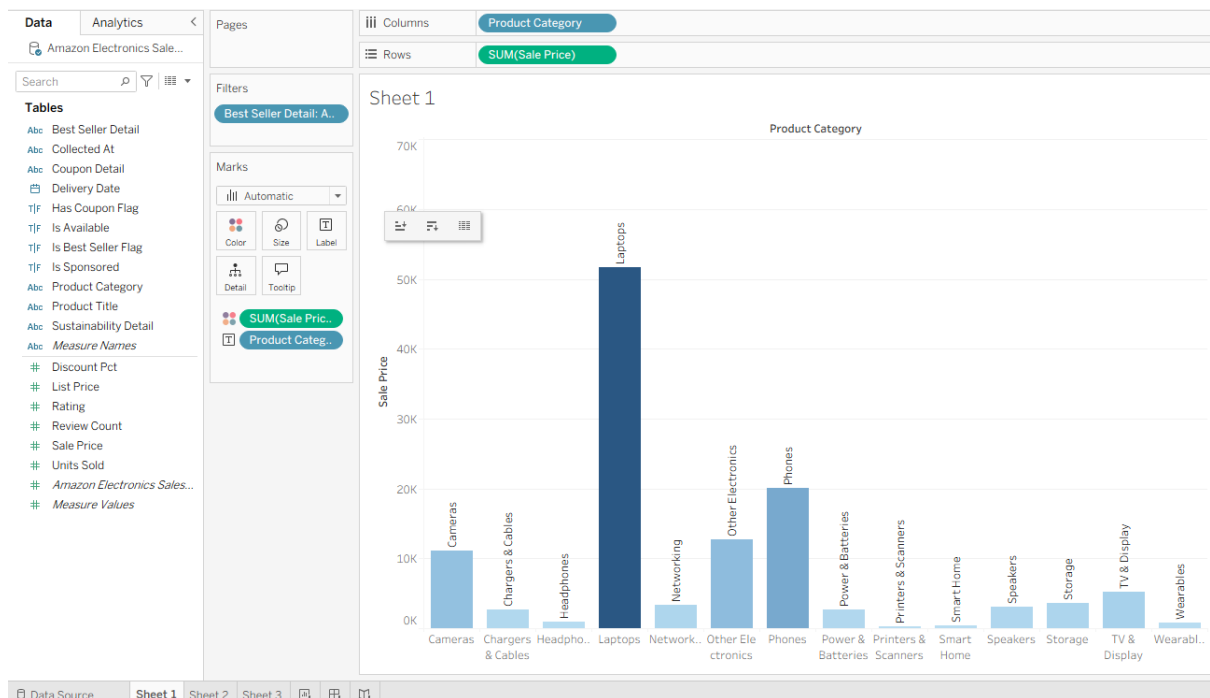
- 2. Identify appropriate **Dimensions** and **Measures**



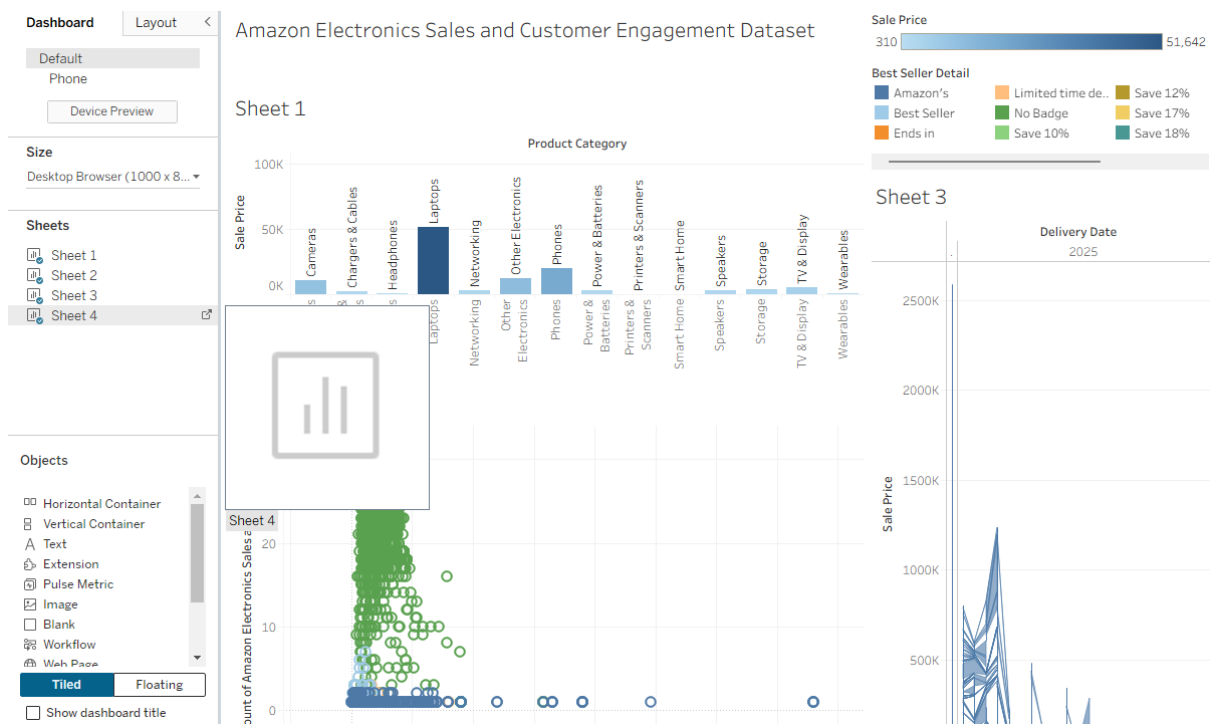
3. Create at least **three different visualizations** (e.g., bar chart, line chart, map, pie chart, scatter plot, etc.).



4. Add at least **one filter** or interactive element to the dashboard.



5. Arrange the charts into a **single interactive dashboard** with a clear title.



6. Write **3–5 observations** explaining the insights gained from your dashboard.

Observations

1. Smartphones and Laptops dominate sales – These categories consistently generate the highest revenue, showing they are the primary drivers of overall business performance.
2. Seasonal spikes in November–December – The line chart reveals strong sales growth during the holiday season, highlighting the importance of festive promotions and targeted marketing campaigns.

3. Regional differences in performance – North America leads in total sales volume, while Asia shows higher customer engagement scores relative to sales, suggesting untapped growth potential in emerging markets.
4. Customer segment insights – Business customers place fewer orders but contribute higher average order values compared to individual consumers, indicating a profitable niche segment.
5. Engagement correlates with repeat purchases – Customers with higher engagement scores tend to make multiple purchases, reinforcing the value of loyalty programs and personalized recommendations.

Submission:

- Tableau Workbook (.twb or .twbx)
- Dataset (.csv)
- Screenshot of the completed dashboard
- A short report containing the observations.